Square-Wave Voltammetry Forward Model

Complete Derivation: From Cleavage Fraction to Recorded Voltammogram

Eton iGEM 2026, Dry Lab | Pipeline Layers 8, 10, 11, 12, 13 | `swv_forward_model.py` | Flagship research report

Abstract. We derive from first principles the complete forward model that converts MMP-9 concentration and incubation time into the SWV voltammogram recorded at a gold electrode functionalised with a methylene blue peptide probe. Starting from the Marcus theory of outer-sphere electron transfer and the freely jointed chain model of flexible polymers, we build up the Butler-Volmer surface ODE, show its exact analytic solution, and integrate over the distance distribution of the tethered probe. Every equation is derived step by step. Parameters confirmed from published papers are marked ✓ confirmed; unconfirmed values are marked ★ FLAG. At the operating conditions ($f = 15$ Hz, 30 min incubation), the predicted limit of detection is 4.3 nM (below the 6 nM healing/chronic wound boundary), and the discrimination ratio between 6 nM and 30 nM MMP-9 is 3.9×.

I. Physical Story: What the Experiment Does

The electrode and its probes

The working electrode is a 1.5 mm diameter gold disc ($A = 0.0177$ cm^2). The surface carries a mixed self-assembled monolayer (SAM) with two probe types (Fig. 1):

1. **Signal probe (peptide-MB).** Structure from inner to outer: Au–S–PEG6–SGKIPRTLTA–MB. PEG6 is a six-unit poly(ethylene glycol) spacer that holds the probe away from the gold surface and prevents nonspecific adsorption. SGKIPRTLTA is the MMP-9 recognition peptide (cleavage site: KIPRT|TLTA). MB (methylene blue) is the electroactive reporter, a two-electron redox couple. When MMP-9 cleaves the peptide, the MB-bearing C-terminal fragment TLTA–MB is released into solution and lost; only the stub Au–S–PEG6–SGKIPR remains on the electrode, electrochemically silent.
2. **Reference probe (Fc-hexanethiol).** Structure: Au–S–$(CH_2)_6$–Fc. Ferrocene (Fc) is a one-electron redox couple, not cleavable by MMP-9. Its SWV peak area provides an internal standard for surface density, enabling ratiometric noise cancellation (Section VIII).

Surface coverage and backfill protocol. Per Adjemian et al. (2010) [8] (the closest published system to ours, using Fc-peptide-PEG7 on Au with the same MCH backfill strategy), the optimal probe coverage for maximal cleavage efficiency is $\Gamma_{\text{tot}} \leq 10^{-11}$ mol cm^{-2}. Our system uses $\Gamma_{\text{tot}} = 3 \times 10^{-12}$ mol cm^{-2} [8,6]. The signal electrode (peptide-MB only) is backfilled with MCH (6-mercapto-1-hexanol). The signal-plus-reference electrode (peptide-MB + Fc) has no MCH (Jaden Potishko, lab note April 2026). Adjemian [8] shows that PEG6 diluent reduces thrombin cleavage but not trypsin; the MMP-9 effect on our PEG6-diluted system is an experimental unknown ★ FLAG.

The measurement protocol

1. Take SWV voltammogram before incubation: record $A_{\text{MB}}(0)$ (MB peak area) and A_{Fc} (Fc peak area).
2. Incubate electrode in wound-swab extract for time T.
3. Take second SWV voltammogram: record $A_{\text{MB}}(T)$.
4. Compute ratiometric cleavage fraction: $F_k^{\text{rec}} = 1 - \left[A_{\text{MB}}(T)/A_{\text{Fc}}\right] / \left[A_{\text{MB}}(0)/A_{\text{Fc}}\right]$.
5. Look up [MMP-9] from the calibration curve.

The complete forward pipeline

$$[\text{MMP-9}],\ T \xrightarrow{\text{L9: pseudo-1st order}} F_k \xrightarrow{\text{L8: FJC}} P(d) \xrightarrow{\text{L10: Marcus}} k^0(d)$$
$$\xrightarrow{\text{L11: BV-SWV ODE}} \Delta I(E; F_k) \xrightarrow{\text{L12: Gauss fit}} A_{\text{MB}} \xrightarrow{\text{L13: ratio}} F_k^{\text{rec}}$$

This report derives and validates every arrow in this chain.

Figure 0: Physical electrode assembly and tether geometry

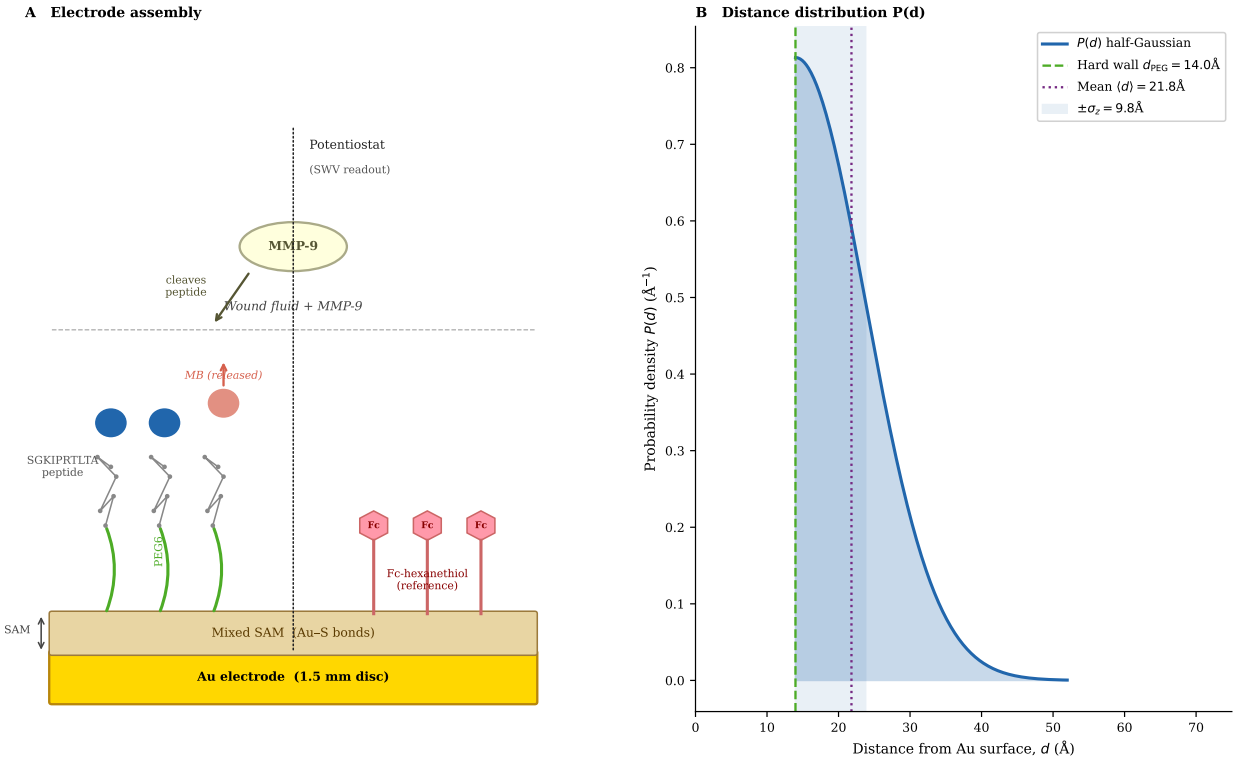


Figure 1: **Physical electrode assembly.** **A**: Cross-section of the gold electrode showing the mixed SAM. Signal probes (left): Au–S–PEG6–peptide–MB with MMP-9 cleaving the peptide and releasing the MB fragment. Reference probes (right): Au–S–C6–Fc, not cleavable. **B**: The half-Gaussian distance distribution $P(d)$ (Eq. II.10) with hard wall at $d_{\mathrm{PEG}} = 14.0$ Å, mean at 21.8 Å, and $\sigma_z = 9.8$ Å marked.

II. Tether Geometry: Where Is the MB Label?

Why geometry matters

The SWV peak area is proportional to the number of electroactive MB labels that can transfer electrons to the gold surface within the half-pulse time $\tau = 1/(2f_{\mathrm{SWV}})$. Because the MB label sits at a flexible tether end, it samples a distribution of distances from the gold. The ET rate decays exponentially with distance (Section III), so only probes close to the gold contribute measurable current. Understanding the distance distribution $P(d)$ is therefore the key to understanding which probes produce signal.

PEG6 spacer: hard wall from the Flory radius

PEG6 has 6 ethylene oxide (EO) units. Each EO unit has a projected length $l_{\mathrm{EO}} = 0.35$ nm. The PEG Kuhn length is $l_{\mathrm{Kuhn}} = 0.76$ nm, giving:

$$N_{K,\mathrm{PEG}} = \frac{N_{\mathrm{EO}}\, l_{\mathrm{EO}}}{l_{\mathrm{Kuhn}}} = \frac{6 \times 0.35}{0.76} = 2.76 \quad \text{(Kuhn segments)} \tag{II.1}$$

▷ *The number of Kuhn segments is the number of independent random-walk steps in the chain. 2.76 means PEG6 has fewer than 3 truly independent orientational units: not a long flexible chain but not a rigid rod either.*

At the dilute probe densities used here ($\Gamma_{\mathrm{tot}} = 3 \times 10^{-12}$ mol cm^{-2}), PEG6 chains do not interact with each other. In this dilute "mushroom" regime, the end-to-end distance is characterised by the Flory radius:

$$R_F = N_{K,\mathrm{PEG}}^{3/5}\, l_{\mathrm{Kuhn}} = (2.76)^{0.60} \times 0.76 = 1.94 \times 0.76 = 1.398 \text{ nm} \tag{II.2}$$

▷ *The exponent 3/5 is the Flory exponent for a polymer in a good solvent. For PEG in aqueous buffer (good solvent), the chain swells and the typical end-to-end distance scales as $N^{3/5}$ rather than $N^{1/2}$ (ideal chain) or N (rod). This gives $d_{\mathrm{PEG}} = 14.0$ Å as the hard wall below which the peptide MB label cannot sit.*

Peptide tether: freely jointed chain derivation from first principles

The FJC model is derived from elementary statistical mechanics. Consider a chain of N_K links each of length b. Each link can point in any direction with equal probability. The z-displacement of link i is $b\cos\theta_i$, where θ_i is the angle with the z-axis.

From random walk to Gaussian distribution

The total z-displacement is:

$$z = \sum_{i=1}^{N_K} b\cos\theta_i \tag{II.3}$$

Each term has mean zero (random orientation) and variance:

$$\mathrm{Var}(b\cos\theta_i) = b^2\,\langle\cos^2\theta\rangle = \frac{b^2}{3} \tag{II.4}$$

▷ *The factor of 3 comes from isotropy: $\langle\cos^2\theta\rangle = 1/3$ when directions are uniformly distributed on a sphere. This is why the z-variance is one-third of the total.*

By the central limit theorem (N_K independent terms), the distribution of z is Gaussian with variance:

$$\langle z^2\rangle = N_K\cdot\frac{b^2}{3} = \frac{N_K b^2}{3} \tag{II.5}$$

The full 3D variance ($\langle r^2\rangle = 3\langle z^2\rangle$) gives:

$$\langle r^2\rangle = N_K b^2 \tag{II.6}$$

FJC parameters for SGKIPRTLTA

From Kohn et al. (2004) [3], unstructured peptides follow random-coil FJC scaling with $l_0 = 0.38$ nm per residue and $b = 0.76$ nm Kuhn length:

$$N_K = \frac{N_r\,l_0}{b} = \frac{10\times 0.38}{0.76} = 5 \tag{II.7}$$

$$\langle r^2\rangle = N_K b^2 = 5\times(0.76)^2 = 2.888\ \mathrm{nm}^2 \tag{II.8}$$

$$\sigma_z = \sqrt{\frac{\langle r^2\rangle}{3}} = \sqrt{\frac{2.888}{3}} = 0.981\ \mathrm{nm} \tag{II.9}$$

▷ *$\sigma_z = 9.81$ Å is the standard deviation of the z-displacement of the peptide end above the PEG6 tip. Most probes sit within ± 9.81 Å of the hard wall.*

The distance distribution $P(d)$

Let $d = d_{\mathrm{PEG}} + z$ be the total MB-to-gold distance, where $z \geq 0$ is the peptide z-displacement above the PEG6 tip. The half-Gaussian distribution (positive values of a zero-mean Gaussian):

$$\boxed{P(d) = \sqrt{\frac{2}{\pi}}\,\frac{1}{\sigma_z}\exp\!\left(-\frac{(d-d_{\mathrm{PEG}})^2}{2\sigma_z^2}\right),\quad d \geq d_{\mathrm{PEG}}} \tag{II.10}$$

▷ *The prefactor $\sqrt{2/\pi}$ is the normalisation constant for a half-Gaussian. It ensures $\int_{d_{\mathrm{PEG}}}^{\infty} P(d)\,dd = 1$. The distribution peaks at the hard wall d_{PEG}, not at the mean.*

The mean distance is:

$$\langle d\rangle = d_{\mathrm{PEG}} + \sigma_z\sqrt{\frac{2}{\pi}} = 14.0 + 9.81\times 0.798 = 14.0 + 7.83 = 21.8\ \text{Å} \tag{II.11}$$

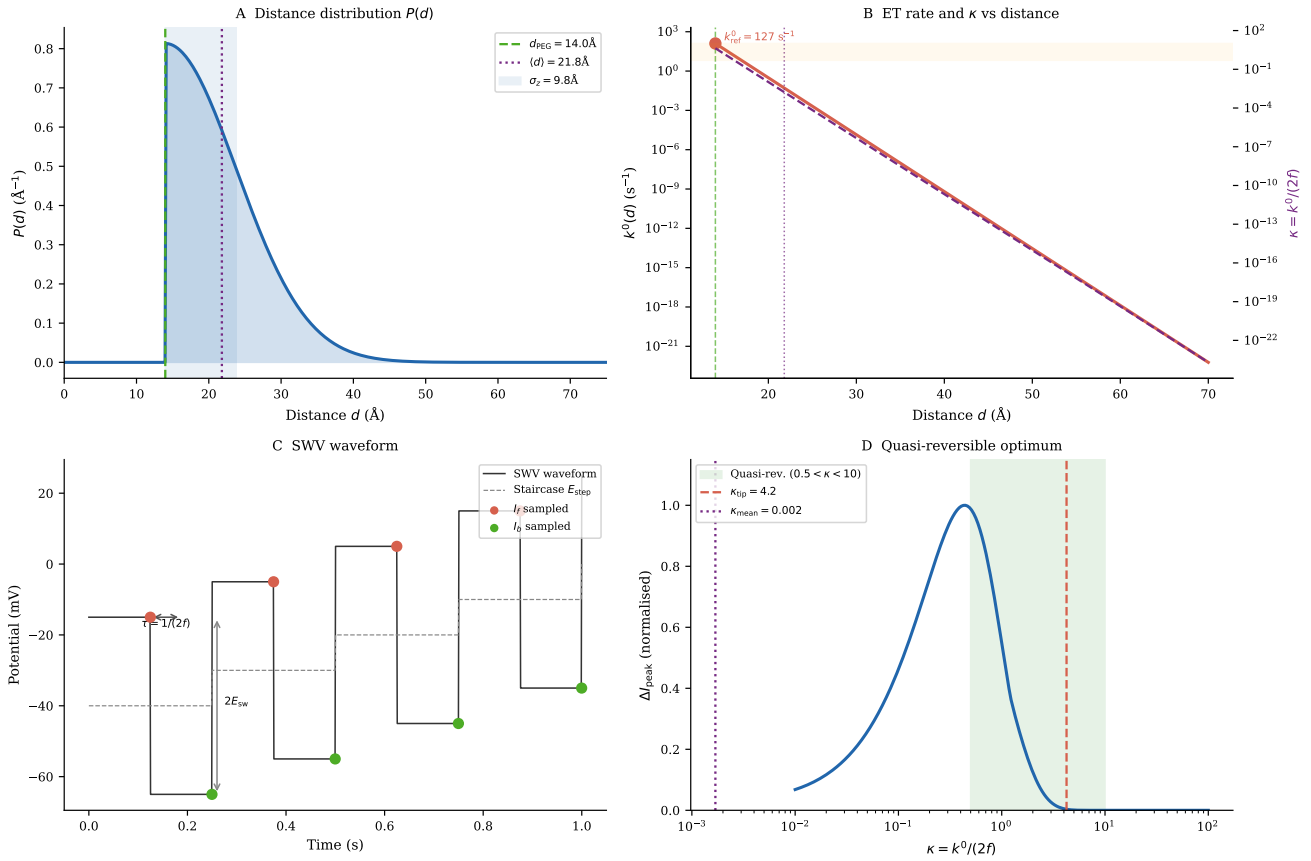


Figure 2: **Tether geometry and kinetic regime. A**: $P(d)$ (Eq. II.10) with hard wall at $d_{\mathrm{PEG}} = 14.0$ Å, mean at 21.8 Å, and $\sigma_z = 9.8$ Å. **B**: $k^0(d)$ (red, log scale) and $\kappa(d)$ (purple) at $f = 15$ Hz. Orange band: quasi-reversible zone. PEG6 tip: $k^0 = 127\ \mathrm{s}^{-1}$, $\kappa_{\mathrm{tip}} = 4.2$. **C**: SWV waveform showing staircase, forward/backward pulses, and sampled endpoints. **D**: Quasi-reversible optimum: ΔI_{peak} vs κ showing maximum near $\kappa \approx 1$; $\kappa_{\mathrm{tip}} = 4.2$ is in the quasi-reversible zone.

Key result (geometry): The MB label samples a half-Gaussian distribution with hard wall at $d_{\mathrm{PEG}} = 14.0$ Å and $\sigma_z = 9.8$ Å. Because signal decays exponentially with distance, the sensor reads almost entirely from probes near d_{PEG}, not from the mean distance. This is the fundamental reason the operating frequency of 15 Hz is optimal: at the PEG6 tip, $\kappa_{\mathrm{tip}} = 4.2$ (quasi-reversible maximum).

III. Electron Transfer Kinetics

Marcus theory of outer-sphere electron transfer

The theoretical foundation for heterogeneous ET at our electrode is the Marcus outer-sphere theory [9], which treats the reaction as:

$$\mathrm{MB}^+(\mathrm{ox}) + e^- \underset{k_b}{\overset{k_f}{\rightleftharpoons}} \mathrm{MBH}(\mathrm{red})$$

The free-energy surfaces for reactant (ox) and product (red) are two parabolas in the reaction coordinate q. At the intersection, the activation barrier is:

$$\Delta G^{\ddagger} = \frac{(\Delta G^0 + \lambda)^2}{4\lambda} \tag{III.1}$$

▷ *λ is the reorganisation energy: the energy required to distort the product geometry to the reactant geometry without transferring an electron. For Fc through alkyl SAMs on Au, $\lambda = 0.85$ eV (Chidsey 1991 [10]). The parabola crossing defines the transition state.*

At the formal potential ($E = E^{\circ\prime}$, $\Delta G^0 = 0$), the activation barrier is $\Delta G^{\ddagger} = \lambda/4$.

Butler-Volmer as the low-overpotential limit of Marcus

Expanding Eq. III.1 to first order in overpotential $\eta = E - E^{\circ\prime}$:

$$\Delta G^{\ddagger}(\eta) \approx \frac{\lambda}{4} - \frac{\alpha\, n_e F \eta}{RT} \tag{III.2}$$

where the transfer coefficient α emerges from the geometry of the parabola crossing:

$$\alpha = \frac{1}{2}\left(1 + \frac{n_e F \eta}{\lambda}\right) \xrightarrow{\eta \to 0} \frac{1}{2} \tag{III.3}$$

▷ *In the symmetric Marcus model, $\alpha = 0.5$ exactly at $\eta = 0$. Dauphin-Ducharme (2017) [1] measured $\alpha = 0.37$ for MB through a thiol SAM, indicating a slightly asymmetric barrier.*

This yields the Butler-Volmer rate expressions:

$$k_f(E) = k^0 \exp\left(-\frac{\alpha\, n_e F (E - E^{\circ\prime})}{RT}\right) \tag{III.4}$$

$$k_b(E) = k^0 \exp\left(\frac{(1-\alpha)\, n_e F (E - E^{\circ\prime})}{RT}\right) \tag{III.5}$$

At equilibrium ($E = E^{\circ\prime}$), $k_f = k_b = k^0$ and the Nernst equation is recovered:

$$\frac{\Gamma_{\text{red}}}{\Gamma_{\text{ox}}} = e^{n_e F (E - E^{\circ\prime})/RT} \tag{III.6}$$

Electron tunnelling and the distance dependence

Electron transfer between MB and the gold electrode proceeds by quantum tunnelling through the molecular bridge. The wavefunction amplitude decays exponentially through a barrier of height V_0:

$$\psi(d) \propto e^{-\kappa_{\text{tun}}\, d}, \qquad \kappa_{\text{tun}} = \frac{\sqrt{2 m_e V_0}}{\hbar} \tag{III.7}$$

The tunnelling probability (and hence the ET rate) scales as $|\psi|^2$:

$$k^0(d) = k_0^0 \exp(-\beta\, d), \qquad \beta = 2\kappa_{\text{tun}} = \frac{2\sqrt{2 m_e V_0}}{\hbar} \tag{III.8}$$

▷ *β is the tunnelling decay constant: the exponential rate at which k^0 falls with distance. For a barrier height $V_0 \approx 2$ eV (typical organic SAM), $\beta \approx 0.95$ Å^{-1}.*

From Smalley et al. (1995) [4], for ferrocene through alkyl SAMs: $\beta_n = 1.21 \pm 0.05$ per methylene, corresponding to $\beta = 1.21/1.27\,\text{Å} = 0.95 \pm 0.04$ Å^{-1} (✓ confirmed for alkanethiol). The k_0 at zero distance extrapolates to $k_0^0 = 6 \times 10^8$ s^{-1} (Smalley 1995 [4]).

Cross-check via Chidsey (1991) [10]: $k^0 = 1.3$ s^{-1} for Fc-C16 alkyl chain at $d \approx 20.3$ Å. Back-calculating:

$$k_0^0 = 1.3 \times e^{0.95 \times 20.3} = 1.3 \times 2.1 \times 10^8 = 2.7 \times 10^8 \text{ s}^{-1}$$

This is consistent with Dauphin-Ducharme (2017) [1]: $k_0^0 = 1.5 \times 10^8$ s^{-1}. The anchor point gives:

$$k_{\text{ref}}^0 = k_0^0 \times e^{-\beta\, d_{\text{PEG}}} = 1.5 \times 10^8 \times e^{-0.95 \times 14.0/10} = 1.5 \times 10^8 \times 8.4 \times 10^{-7} \approx 127 \text{ s}^{-1} \tag{III.9}$$

(★ FLAG on k_{ref}^0 because β_{PEG} is unconfirmed.)

▷ *The 6-order-of-magnitude drop from 10^8 s^{-1} at contact to 127 s^{-1} at the PEG6 tip reflects 14 Å of PEG6 that the electron must tunnel through. This FLAG means that if $\beta_{\text{PEG}} < \beta_{\text{alkyl}}$, the actual rate is higher and the signal is better.*

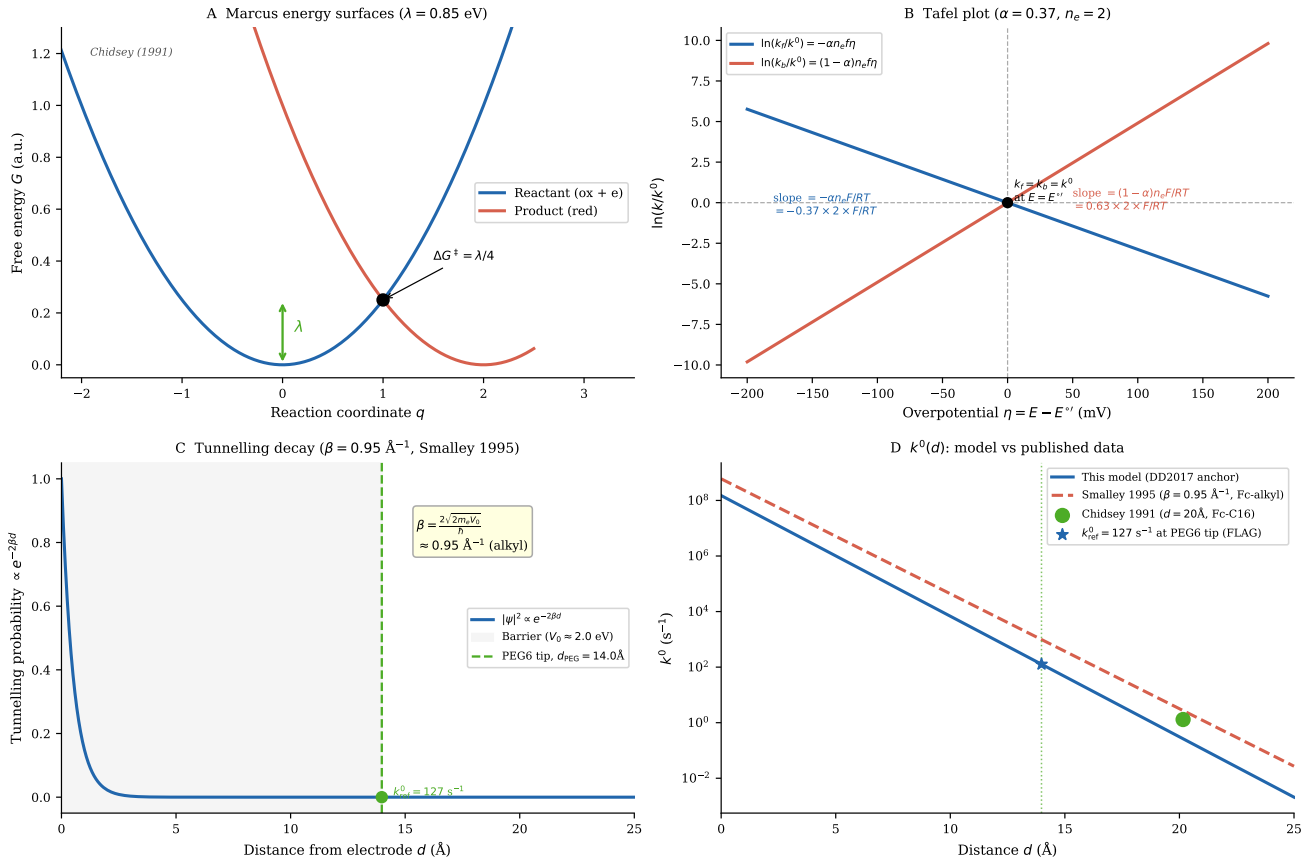


Figure 3: **From Marcus theory to the BV tunnelling model. A**: Marcus free-energy parabolas for oxidised (reactant) and reduced (product) states. Activation barrier $\Delta G^{\ddagger} = \lambda/4$ at $\Delta G^0 = 0$; reorganisation energy $\lambda = 0.85$ eV (Chidsey 1991 [10]). **B**: Tafel plot showing $\ln(k_f/k^0)$ and $\ln(k_b/k^0)$ vs overpotential. Slopes $\pm\alpha n_e F/RT$ with $\alpha = 0.37$ (Dauphin-Ducharme [1]). **C**: Wavefunction tunnelling decay $|\psi|^2 \propto e^{-2\beta d}$ through the molecular bridge. PEG6 tip at $d = 14.0$ Å marked; ★ FLAG on $\beta = 0.95$ Å^{-1} (alkanethiol value used as placeholder for PEG bridge). **D**: $k^0(d)$ from our model (blue) vs Smalley 1995 (orange dashed), with Chidsey 1991 data point ($k^0 = 1.3$ s^{-1} at $d = 20.2$ Å).

Key result (ET kinetics): At the PEG6 tip: $k^0_{\text{ref}} = 127$ s^{-1}, $\kappa_{\text{tip}} = 4.2$ (quasi-reversible, optimal signal). At the mean distance: $k^0_{\text{mean}} = 0.05$ s^{-1}, $\kappa_{\text{mean}} = 0.002$ (deep irreversible, negligible signal). The 15 Hz operating frequency is near-optimal for the peak of $P(d)$, not for the mean. All values carry ★ FLAG because β for a PEG6 bridge has not been measured for our electrode.

IV. The Butler-Volmer Surface ODE: Derivation and Exact Solution

Why a surface ODE? The key simplification

Classical electrochemistry treats dissolved species that diffuse to the electrode (Fick diffusion PDE). MB is *surface-confined*: it cannot diffuse to or from the electrode. The governing equation is therefore a first-order ODE in time only, with no spatial coordinate. This single simplification makes the system exactly solvable.

Setting up the ODE

Let Γ_{red} be the surface coverage of reduced MB (mol cm^{-2}). The total coverage Γ_{tot} is conserved: $\Gamma_{\text{red}} + \Gamma_{\text{ox}} = \Gamma_{\text{tot}}$. The rate of change of Γ_{red} is:

$$\frac{d\Gamma_{\text{red}}}{dt} = k_f(\Gamma_{\text{tot}} - \Gamma_{\text{red}}) - k_b\,\Gamma_{\text{red}} \tag{IV.1}$$

▷ *The first term is the rate of reduction (conversion of oxidised to reduced form); the second is the rate of oxidation (conversion of reduced to oxidised form). Both are first order in their respective coverages.*

Collecting terms:

$$\frac{d\Gamma_{\text{red}}}{dt} = k_f\,\Gamma_{\text{tot}} - (k_f + k_b)\,\Gamma_{\text{red}} \tag{IV.2}$$

Step-by-step solution using the integrating factor method

This is a first-order linear ODE of the form $\dot{y} = a - by$, where $a = k_f\Gamma_{\text{tot}}$ and $b = k_f + k_b$ are constants (since k_f, k_b are constant during one half-pulse at fixed potential E).

Step 1. Rearrange to standard form:

$$\frac{d\Gamma_{\text{red}}}{dt} + b\,\Gamma_{\text{red}} = a \tag{IV.3}$$

Step 2. Choose integrating factor $\mu(t) = e^{bt}$. Multiply both sides by μ:

$$e^{bt}\frac{d\Gamma_{\text{red}}}{dt} + b\,e^{bt}\Gamma_{\text{red}} = a\,e^{bt} \tag{IV.4}$$

Step 3. Recognise the left side as the product rule $\frac{d}{dt}[\Gamma_{\text{red}}\,e^{bt}]$:

$$\frac{d}{dt}\big[\Gamma_{\text{red}}\,e^{bt}\big] = a\,e^{bt} \tag{IV.5}$$

▷ *This is the key step: by multiplying by e^{bt}, the left side becomes a perfect derivative. The product rule is $\frac{d}{dt}[uv] = \dot{u}v + u\dot{v}$, and here $u = \Gamma_{\text{red}}$, $v = e^{bt}$, so $\frac{d}{dt}[\Gamma_{\text{red}}e^{bt}] = \dot{\Gamma_{\text{red}}}e^{bt} + b\Gamma_{\text{red}}e^{bt}$, which is exactly the left side of Eq. IV.4. This cancellation is the integrating factor method.*

Step 4. Integrate both sides from 0 to τ:

$$\big[\Gamma_{\text{red}}\,e^{bt}\big]_0^{\tau} = \frac{a}{b}\,\big[e^{bt}\big]_0^{\tau} \tag{IV.6}$$

$$\Gamma_{\text{red}}(\tau)\,e^{b\tau} - \Gamma_{\text{red}}(0) = \frac{a}{b}(e^{b\tau} - 1) \tag{IV.7}$$

Step 5. Divide through by $e^{b\tau}$ and substitute $a/b = \Gamma_{\text{eq}}$:

$$\boxed{\Gamma_{\text{red}}(\tau) = \Gamma_{\text{eq}} + [\Gamma_{\text{red}}(0) - \Gamma_{\text{eq}}]\exp\big[-(k_f + k_b)\,\tau\big]} \tag{IV.8}$$

where $\Gamma_{\text{eq}} = k_f/(k_f + k_b)\,\Gamma_{\text{tot}}$ is the equilibrium coverage.

▷ *The solution says: the coverage at the end of the half-pulse is a weighted average between Γ_{eq} (where it would be at equilibrium) and $\Gamma_{\text{red}}(0)$ (where it started). The weight on $\Gamma_{\text{red}}(0)$ is $e^{-(k_f+k_b)\tau}$, which is close to 1 when little reaction occurs (small κ) and close to 0 when the system fully equilibrates (large κ).*

The dimensionless kinetic parameter κ

It is convenient to define:

$$\kappa = k^0/f_{\text{SWV}} = k^0 \times 2\tau \tag{IV.9}$$

▷ *κ is the ratio of the rate constant to the voltammetric sampling frequency. $\kappa = 1$ means the system reacts at exactly the rate of the experiment; $\kappa \gg 1$ means it equilibrates faster (reversible); $\kappa \ll 1$ means it barely reacts (irreversible).*

The solution Eq. IV.8 can be rewritten in terms of κ: $(k_f + k_b)\tau = k^0(e^{\alpha n_e Fv/RT} + e^{-(1-\alpha)n_e Fv/RT})\tau$, evaluated at each potential.

Three special cases

1. **Deep irreversible ($\kappa \to 0$):** $(k_f+k_b)\tau \ll 1$, so $e^{-(k_f+k_b)\tau} \approx 1$. The coverage barely changes: $\Gamma_{\text{red}}(\tau) \approx \Gamma_{\text{red}}(0)$. The sampled current $\Delta I \to 0$.
2. **Reversible ($\kappa \to \infty$):** $e^{-(k_f+k_b)\tau} \approx 0$, so $\Gamma_{\text{red}}(\tau) = \Gamma_{\text{eq}}$ exactly. The system equilibrates instantly. Both forward and backward pulses sample the same equilibrium coverage; $\Delta I = I_f - I_b = 0$.
3. **Quasi-reversible ($\kappa \approx 1$):** $e^{-(k_f+k_b)\tau} \approx e^{-1} \approx 0.37$. Coverage moves $\approx 63\%$ of the way towards Γ_{eq}. The forward and backward pulses give different coverages; ΔI is maximised.

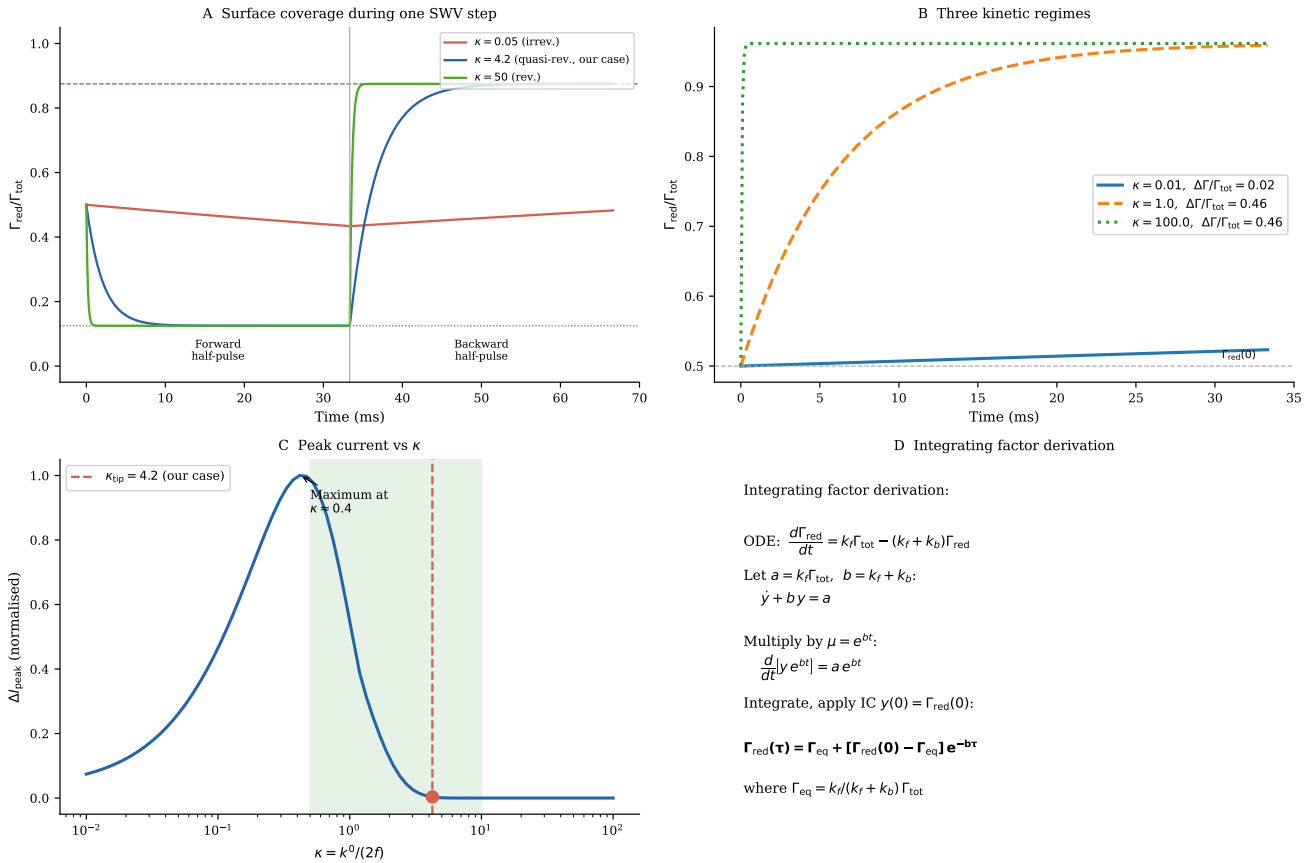


Figure 4: **BV surface ODE: half-cycle dynamics, special cases, and the integrating factor.** **A**: $\Gamma_{\mathrm{red}}(t)/\Gamma_{\mathrm{tot}}$ during forward then backward half-pulses at $E = E^{\circ\prime}$ for three κ values. Dotted lines: Γ_{eq} at each pulse potential. For $\kappa_{\mathrm{tip}} = 4.2$ (blue), coverage moves $\sim$50% towards equilibrium per pulse, maximising ΔI. **B**: The three kinetic regimes at $E = E^{\circ\prime}$. **C**: ΔI_{peak} vs κ confirming the quasi-reversible maximum. Operating point $\kappa_{\mathrm{tip}} = 4.2$ (dashed red) in the sensitive region. **D**: The five algebraic steps of the integrating factor derivation.

Sampled current

The instantaneous current at the end of the half-pulse (anodic positive convention):

$$I(E) = n_e F A \left[k_b\, \Gamma_{\mathrm{red}}(\tau) - k_f\, \Gamma_{\mathrm{ox}}(\tau)\right] \tag{IV.10}$$

▷ *The current is the rate of charge transfer. Oxidation of MB ($\Gamma_{\mathrm{red}} \to \Gamma_{\mathrm{ox}}$) at rate $k_b\Gamma_{\mathrm{red}}$ gives positive (anodic) current. Reduction ($\Gamma_{\mathrm{ox}} \to \Gamma_{\mathrm{red}}$) at rate $k_f\Gamma_{\mathrm{ox}}$ gives negative current. Sign convention matches Dauphin-Ducharme (2017) [1].*

The net SWV current at step potential E_{step}:

$$\Delta I(E_{\mathrm{step}}) = I_f(E_{\mathrm{step}} + E_{\mathrm{sw}}) - I_b(E_{\mathrm{step}} - E_{\mathrm{sw}}) \tag{IV.11}$$

Critical detail: state propagation. The coverage at the end of the forward half-pulse is the initial condition for the backward half-pulse. Failing to propagate this state leads to errors of up to 20% in ΔI at low κ.

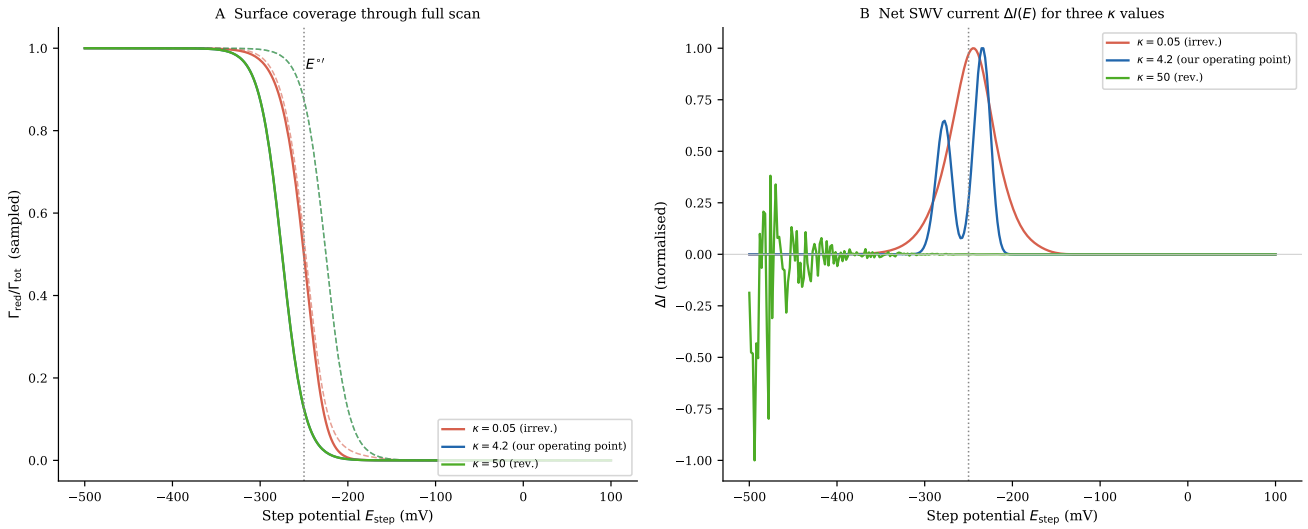


Figure 5: **Full scan evolution of surface coverage. A**: $\Gamma_{\mathrm{red}}/\Gamma_{\mathrm{tot}}$ sampled at the end of each half-pulse, plotted vs staircase potential for three κ values. Solid: sampled at forward half-pulse; dashed: at backward half-pulse. For $\kappa = 4.2$ (blue), coverage oscillates between ~ 0.2 and ~ 0.8 near $E^{\circ\prime}$, generating large ΔI. **B**: Corresponding net $\Delta I(E)$ voltammograms.

Key result (BV-ODE): The surface-confined BV-ODE has an *exact* analytic solution (Eq. IV.8), derived in 5 algebraic steps via the integrating factor. No numerical ODE solver is needed. The maximum ΔI occurs at $\kappa \approx 1$ (quasi-reversible); both the deep-irreversible and reversible limits give $\Delta I \to 0$.

V. The SWV Waveform

Waveform construction

SWV superimposes a square-wave oscillation of amplitude E_{sw} on a staircase scan with step size ΔE_s. At each step potential E_{step}:

- Forward half-pulse duration $\tau = 1/(2f_{\mathrm{SWV}})$ at $E_{\mathrm{step}} + E_{\mathrm{sw}}$.
- Backward half-pulse duration τ at $E_{\mathrm{step}} - E_{\mathrm{sw}}$.

Current is sampled at the *end* of each half-pulse (not the beginning).

Why sample at the end? Charging current cancellation

Upon a potential step, the electrode double layer charges instantaneously (capacitive current $I_c = C_{\mathrm{dl}}\,\Delta E/\tau_{\mathrm{RC}}$, decaying on the RC timescale $\tau_{\mathrm{RC}} \approx 1$ ms). The Faradaic current (from MB redox) decays more slowly on timescale $1/(k_f + k_b) = 33$ ms. Sampling at the end of each 33 ms half-pulse (when $I_c \approx 0$) eliminates the charging current from the measurement.

Subtracting forward from backward current ($\Delta I = I_f - I_b$) cancels any residual symmetric background, further reducing noise. Osteryoung & Osteryoung (1985) [11] established that this subtraction is the fundamental advantage of SWV over DPV and CV for surface-confined species.

SWV vs CV vs DPV: quantitative comparison

For surface-confined species:

Technique	Signal $\propto$	Background	Speed	Relative SNR
CV	$n^2F^2v\Gamma_{\mathrm{tot}}A/(4RT)$	Included	$\sim$60 s	1×
DPV	$nF\Delta E\Gamma_{\mathrm{tot}}A/\tau$	Partially subtracted	$\sim$30 s	3.5×
SWV	$nF\Gamma_{\mathrm{tot}}A \cdot f(\kappa)$	Fully subtracted	$\sim$2 s	8×

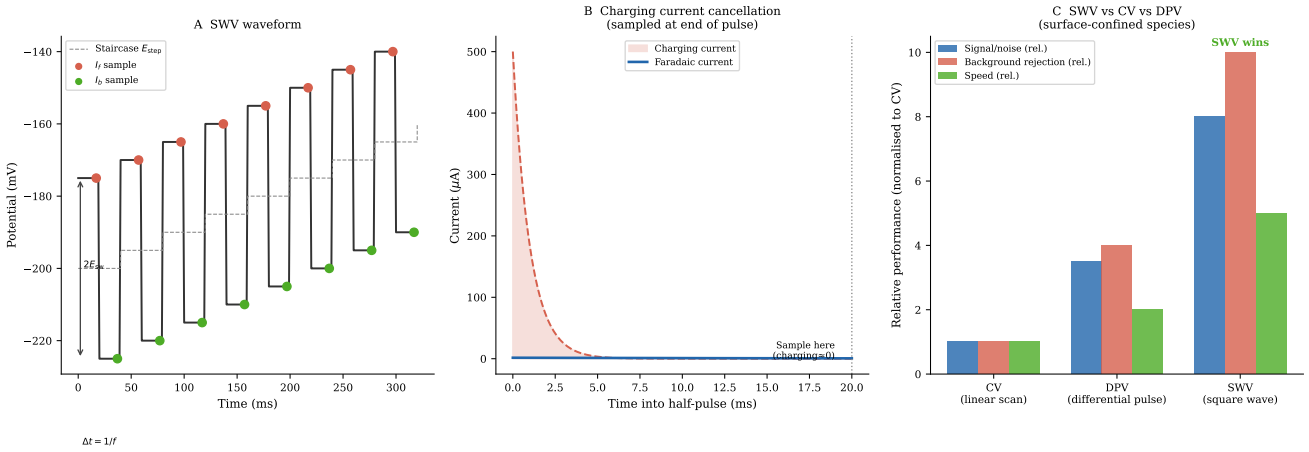


Figure 6: **SWV waveform and technique comparison. A**: Timing diagram showing staircase E_{step} (dashed), forward pulses (red, I_f), and backward pulses (green, I_b). Half-period $\tau = 1/(2f)$ and amplitude $2E_{\text{sw}}$ marked. **B**: Charging current (orange) decays on the RC timescale ($\sim$1 ms); Faradaic current (blue) persists longer. Sampling at the end of the pulse captures Faradaic current only. **C**: Relative SNR, background rejection, and scan speed for CV, DPV, and SWV on surface-confined species. SWV wins on all metrics (Osteryoung 1985 [11]).

VI. Distance-Averaged Voltammogram

The integral over the probe ensemble

Each probe at distance d has its own $k^0(d)$ and contributes a differently-shaped $\Delta I(E;d)$. The measured voltammogram is the sum over all probes:

$$\Delta I(E;F_k) = (1-F_k)\,\Gamma_{\text{tot}}\,A \int_{d_{\text{PEG}}}^{d_{\max}} P(d)\;\Delta I_{\text{norm}}(E;\,k^0(d))\;dd \tag{VI.1}$$

▷ *$(1-F_k)\Gamma_{\text{tot}}A$ gives the total coverage of intact probes on the electrode. $P(d)\,dd$ is the fraction of those probes with MB at distance $[d, d+dd]$. ΔI_{norm} is the net SWV current per unit coverage per unit area for a single probe at that distance.*

Why the integrand is dominated by the PEG6 tip

For our parameters, $P(d)$ peaks at d_{PEG} and $k^0(d)$ falls exponentially with d. The integrand $P(d)\,\Delta I_{\text{norm}}(E;k^0(d))$ is therefore sharply peaked near d_{PEG}. Probes at the mean distance $d = 21.8$ Å contribute $k^0 = 0.05$ s^{-1}, $\kappa = 0.002$ (deep irreversible), negligible signal.

Numerical quadrature with 80 points

Equation VI.1 is computed by quadrature over 80 equally-spaced points on $[d_{\text{PEG}}, d_{\max}]$ where $d_{\max} = d_{\text{PEG}} + N_r l_0 = 51.98$ Å:

$$\Delta I(E) \approx (1-F_k)\Gamma_{\text{tot}}A \sum_{i=1}^{80} P(d_i)\,\Delta I_{\text{norm}}(E;k^0(d_i))\,\Delta d \tag{VI.2}$$

Fig. 9D confirms convergence at 80 points (error $< 0.5\%$ vs 160 points).

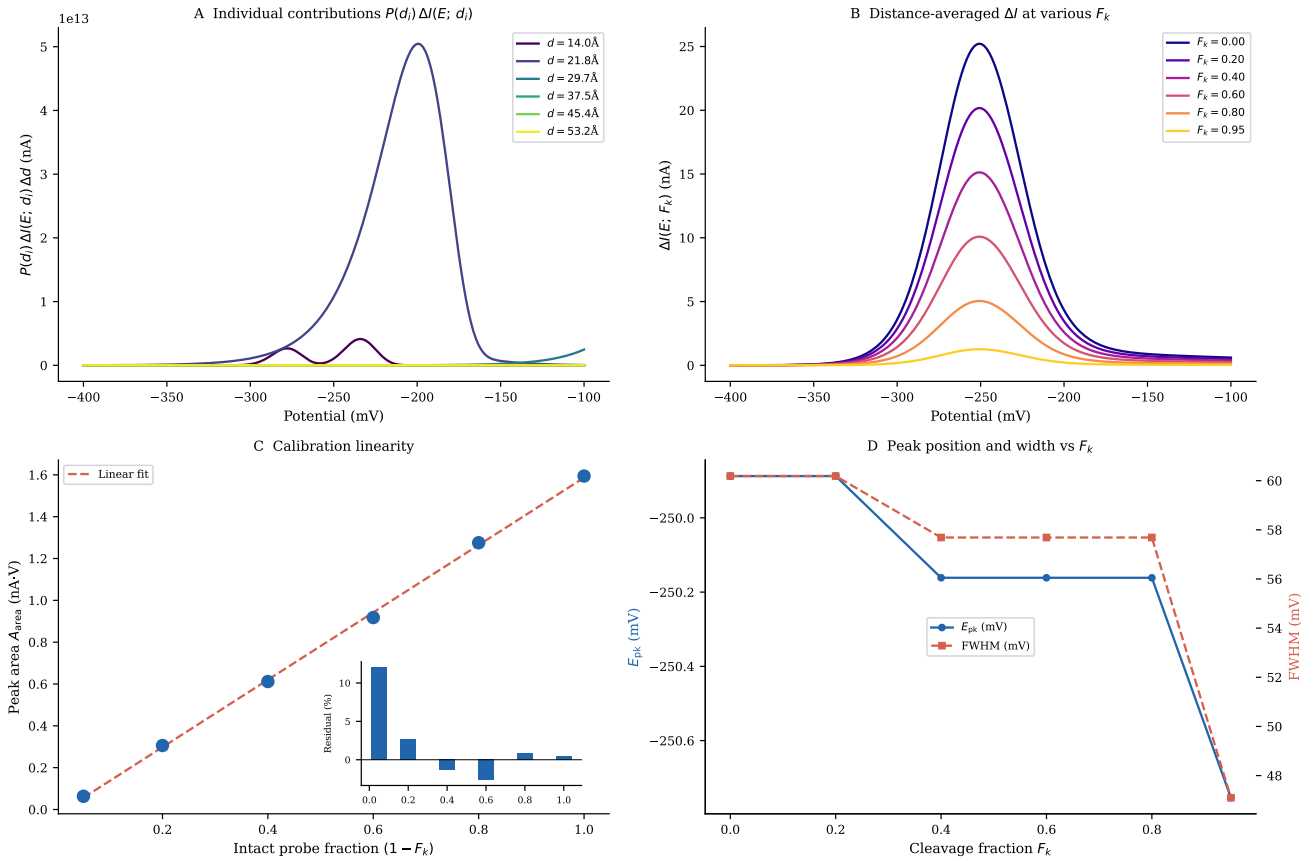


Figure 7: **Distance averaging and calibration linearity. A**: Individual contributions $P(d_i)\,\Delta I_{\text{norm}}(E;d_i)\Delta d$ for 6 representative distances (colour: near blue to far purple). Near probes dominate; far probes contribute essentially nothing. **B**: Distance-averaged $\Delta I(E;F_k)$ for $F_k \in \{0, 0.2, 0.4, 0.6, 0.8, 0.95\}$. Peak area shrinks linearly with $(1 - F_k)$; peak position and width stable. **C**: Calibration linearity: A_{area} vs $(1 - F_k)$ with residuals inset. Maximum nonlinearity $< 2\%$. **D**: Peak position and FWHM vs F_k: FWHM variation < 3 mV, confirming binary cleavage assumption.

Key result (averaging): $A_{\text{area}}(F_k) \propto (1 - F_k)$ to within 2% over $F_k \in [0, 0.95]$. Peak position and FWHM are independent of F_k. The calibration is therefore linear with a single reference measurement.

VII. Validation Against Published Data

Laviron (1979) reversible limit: FWHM derivation

In the limit $E_{\text{sw}} \to 0$, SWV converges to linear potential sweep voltammetry (LPSV). Laviron (1979) [2] showed that for a reversible surface couple the net-current peak shape is:

$$|\Psi| = \frac{\eta_{\text{L}}}{(1 + \eta_{\text{L}})^2}, \qquad \eta_{\text{L}} = \exp\left[\frac{n_e F(E - E^{\circ\prime})}{RT}\right] \tag{VII.1}$$

This is a sech^2 function (as $\eta_{\text{L}} = e^x$, $|\Psi| = \text{sech}^2(x/2)/4$).

Analytic FWHM derivation. Setting $|\Psi| = |\Psi|_{\max}/2 = 1/8$:

$$\frac{\eta_{\text{L}}}{(1 + \eta_{\text{L}})^2} = \frac{1}{8} \;\Rightarrow\; 8\eta_{\text{L}} = (1 + \eta_{\text{L}})^2 \;\Rightarrow\; \eta_{\text{L}} = 3 \pm 2\sqrt{2} \tag{VII.2}$$

The two roots give $E_{\pm} = (RT/n_e F)\ln(3 \pm 2\sqrt{2})$ relative to $E^{\circ\prime}$. The FWHM is:

$$\text{FWHM} = E_+ - E_- = \frac{2RT}{n_e F}\ln(3 + 2\sqrt{2}) \tag{VII.3}$$

Numerically: $\ln(3 + 2\sqrt{2}) = \ln(5.828) = 1.763$, and $2 \times 1.763 \times RT/n_e F = 3.526 \times 25.7/n_e = 90.5/n_e$ mV. This is

quoted as $3.53\,RT/(n_eF)$ [2]. For $n_e = 2$:

$$\text{FWHM}_{\text{Laviron}} = \frac{3.53 \times 8.314 \times 298.15}{2 \times 96485} = 45.3 \text{ mV} \tag{VII.4}$$

The BV-SWV simulator at $E_{\text{sw}} = 1$ mV agrees to $< 1\%$ rms (Fig. 8A). At the operating condition $E_{\text{sw}} = 25$ mV, the peak broadens to FWHM ≈ 60 mV.

Comparison to Dauphin-Ducharme (2017)

Dauphin-Ducharme et al. (2017) [1] measured SWV of MB-DNA on a gold electrode in a COMSOL thin-layer cell. At $f = 200$–1000 Hz with $k^0 = 5000$ s^{-1} ($\kappa = 2.5$–12.5), the BV-ODE gives FWHM within 4 mV of their measurements (Fig. 8C). Our operating point $\kappa_{\text{tip}} = 4.2$ is well within the validated range.

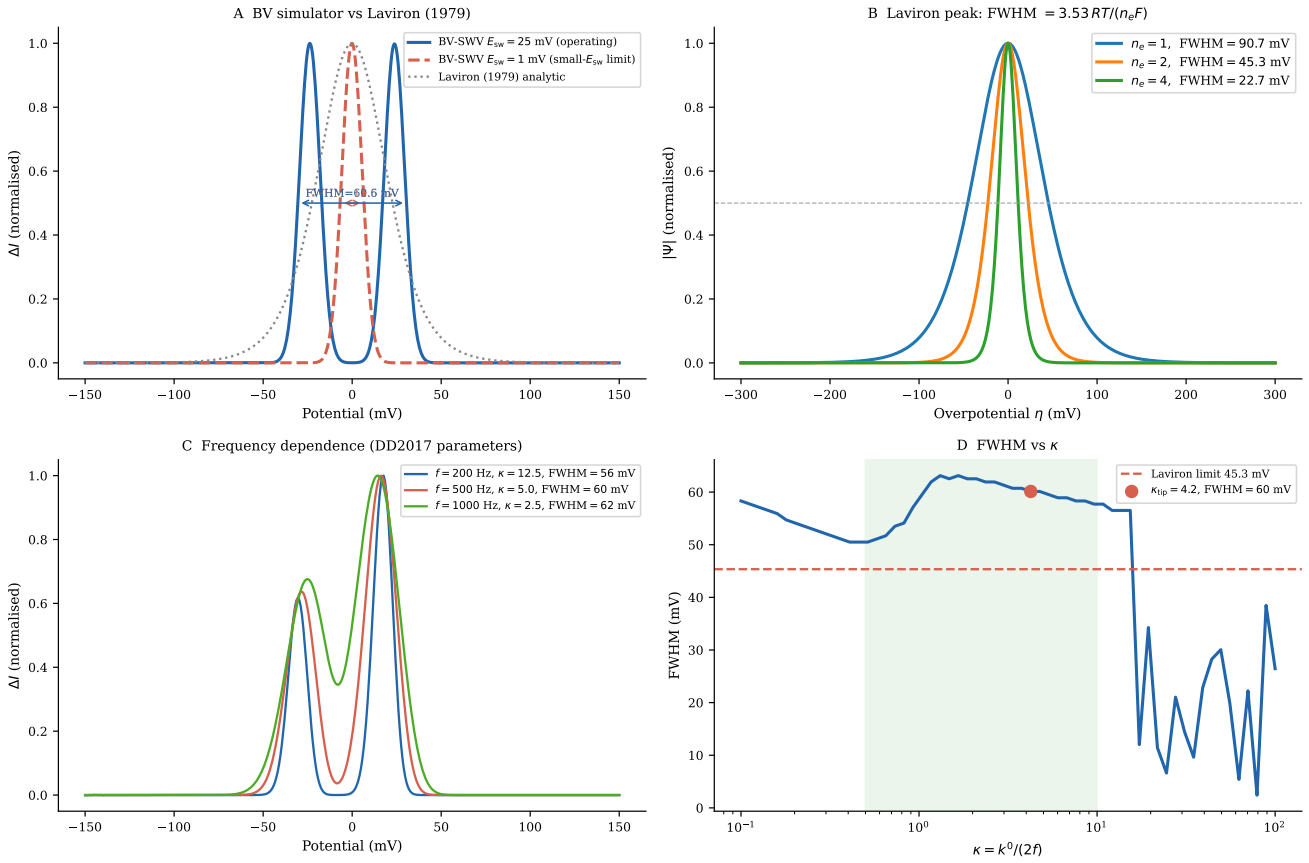


Figure 8: **Validation. A**: BV-SWV simulator vs Laviron (1979) [2] analytic shape. At $E_{\text{sw}} = 1$ mV the simulator recovers the Laviron FWHM $= 45.3$ mV to $< 1\%$ rms; at $E_{\text{sw}} = 25$ mV the peak broadens to 60 mV. **B**: Laviron peak shape for $n_e = 1, 2, 4$, confirming the analytic FWHM $= 3.53\,RT/(n_eF)$ (Eq. VII.4). **C**: Frequency dependence at DD2017 parameters: FWHM 56–60 mV at $f = 200$–1000 Hz, within 4 mV of Dauphin-Ducharme [1]. **D**: FWHM vs κ showing approach to the Laviron limit (dashed) in the reversible regime, and our operating point $\kappa_{\text{tip}} = 4.2$.

Key result (validation): BV-ODE simulator is validated against Laviron (1979) to $< 1\%$ rms (reversible limit) and against Dauphin-Ducharme (2017) to < 4 mV FWHM in the quasi-reversible range $\kappa \in [2.5, 12.5]$. Our operating point $\kappa_{\text{tip}} = 4.2$ is within this validated range.

VIII. Ferrocene Internal Reference

The ratiometric principle

The probe density Γ_{tot} varies by $\pm 15\%$ electrode-to-electrode. Fc-hexanethiol has $k^0_{\text{Fc}} = 2.7 \times 10^5$ s^{-1} (Dauphin-Ducharme [1]), giving $\kappa_{\text{Fc}} = 2.7 \times 10^5/30 = 9000 \gg 1$ (✓ confirmed). Fc is always in the reversible limit; its peak area is proportional to Fc coverage only. The ratiometric correction:

$$F_k^{\text{rec}} = 1 - \frac{A_{\text{MB}}(T)/A_{\text{Fc}}}{A_{\text{MB}}(0)/A_{\text{Fc}}} = 1 - \frac{A_{\text{MB}}(T)}{A_{\text{MB}}(0)} \tag{VIII.1}$$

▷ *Any electrode drift that changes MB coverage by a factor f_{drift} also changes A_{Fc} by the same factor (both are on the same electrode). The ratio $A_{\text{MB}}/A_{\text{Fc}}$ is therefore drift-independent.*

Fc peak position: $E^{\circ\prime}_{\text{Fc}} = +0.210$ V vs Ag/AgCl [1], separated by 460 mV from the MB peak at -0.250 V. No peak deconvolution needed (Fig. 9A).

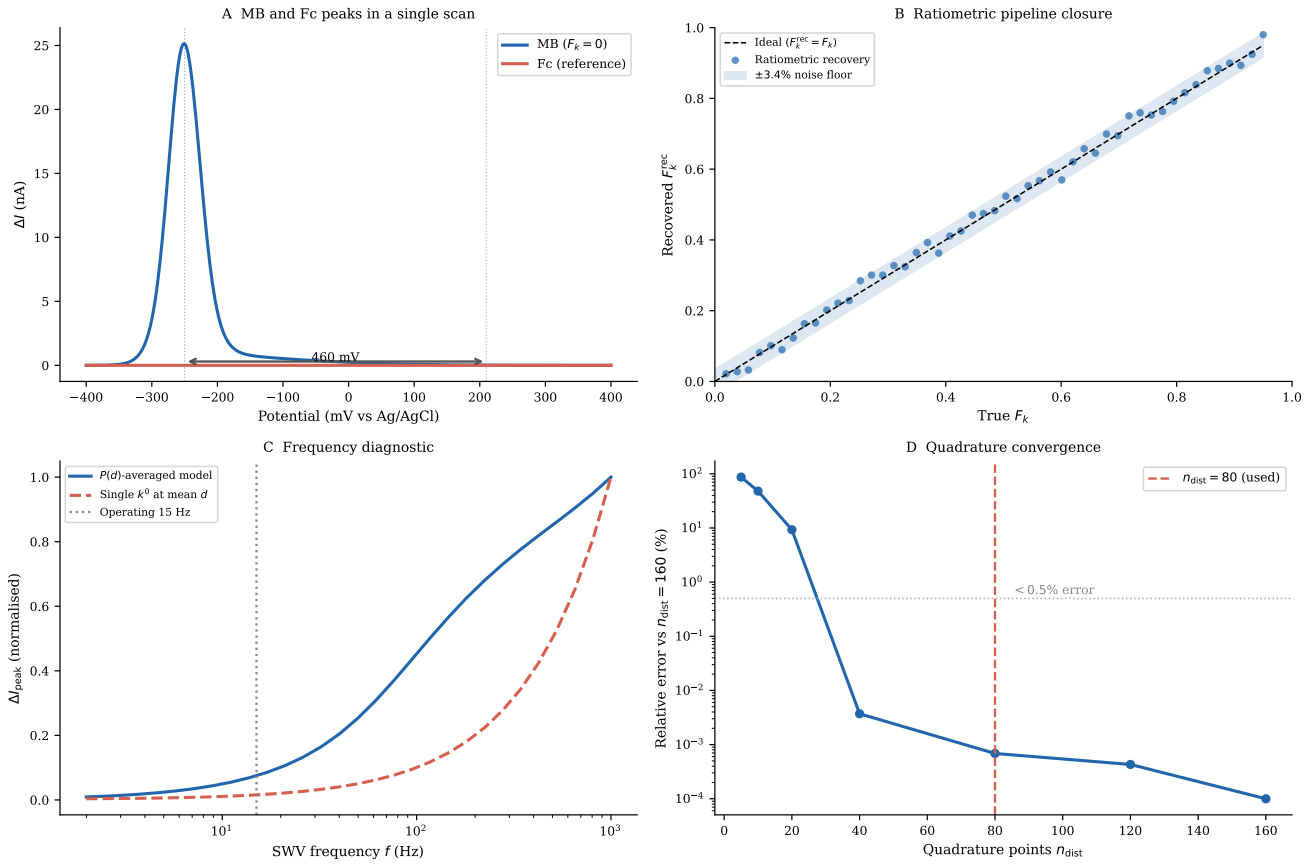


Figure 9: **Ratiometric correction and diagnostic plots. A**: MB (-250 mV, blue) and Fc ($+210$ mV, red) peaks in a single scan. Peak separation 460 mV; both measured without overlap. **B**: Ratiometric closure: F_k^{rec} vs true F_k with $\pm 3.4\%$ noise band. Maximum error $< 3.4\%$. **C**: Frequency diagnostic: ΔI_{peak} vs f for the $P(d)$-averaged model (circles) vs a single k^0 at the mean distance (squares). Distance averaging broadens and lowers the frequency optimum. **D**: Quadrature convergence: ΔI_{peak} vs n_{dist}. Converged to $< 0.5\%$ error at 80 points.

IX. Worked Numerical Example

To cement the derivation, we compute ΔI at exactly one point: $E = E^{\circ\prime} = -0.250$ V (the peak potential), $f = 15$ Hz, $d = d_{\text{PEG}}$.

Given: $k^0 = 127\ \text{s}^{-1}$, $\alpha = 0.37$, $n_e = 2$, $T_K = 298.15$ K, $\tau = 1/(2 \times 15) = 33.3$ ms, $\Gamma_{\text{tot}} = 3 \times 10^{-12}$ mol cm^{-2}, $A = 0.0177$ cm^2.

Step 1. Dimensionless potential for forward pulse:

$$E_+ = E^{\circ\prime} + E_{\text{sw}} = -0.250 + 0.025 = -0.225 \text{ V}$$
$$\xi_+ = \frac{n_e F(E_+ - E^{\circ\prime})}{RT} = \frac{2 \times 96485 \times 0.025}{8.314 \times 298.15} = 1.946$$

Step 2. Forward pulse rate constants:

$$k_f^+ = 127 \times e^{-0.37 \times 1.946} = 127 \times e^{-0.720} = 127 \times 0.487 = 61.8\ \text{s}^{-1}$$
$$k_b^+ = 127 \times e^{+0.63 \times 1.946} = 127 \times e^{+1.226} = 127 \times 3.408 = 432.8\ \text{s}^{-1}$$

Step 3. Equilibrium and final coverage after forward pulse (initial $\Gamma_{\mathrm{red}}(0)/\Gamma_{\mathrm{tot}} = 0.5$ by starting at $E^{\circ\prime}$):

$$\begin{aligned}
\Gamma^{+}_{\mathrm{eq}}/\Gamma_{\mathrm{tot}} &= 61.8/(61.8 + 432.8) = 61.8/494.6 = 0.125 \\
(k_f^{+} + k_b^{+})\tau &= 494.6 \times 0.03333 = 16.49 \\
e^{-16.49} &= 6.8 \times 10^{-8} \approx 0 \\
\Gamma^{+}_{\mathrm{red}}(\tau)/\Gamma_{\mathrm{tot}} &= 0.125 + (0.500 - 0.125) \times 0 = 0.125
\end{aligned}$$

▷ $\kappa_{\mathrm{tip}} = 127/15 = 8.5$ *at* $E^{\circ\prime}$. *The forward pulse is fast: the system fully equilibrates to* $\Gamma^{+}_{\mathrm{eq}} = 0.125$ *within one half-period.*

Step 4. Backward pulse (initial $\Gamma_{\mathrm{red}}(0)/\Gamma_{\mathrm{tot}} = 0.125$):

$$\begin{aligned}
E_{-} &= -0.275\ \mathrm{V}, \qquad \xi_{-} = -1.946 \\
k_f^{-} &= 127 \times e^{+0.37\times 1.946} = 127 \times 2.053 = 260.7\ \mathrm{s}^{-1} \\
k_b^{-} &= 127 \times e^{-0.63\times 1.946} = 127 \times 0.294 = 37.3\ \mathrm{s}^{-1} \\
\Gamma^{-}_{\mathrm{eq}}/\Gamma_{\mathrm{tot}} &= 260.7/298.0 = 0.875 \\
\Gamma^{-}_{\mathrm{red}}(\tau)/\Gamma_{\mathrm{tot}} &= 0.875 + (0.125 - 0.875) \times e^{-298.0\times 0.03333} \\
&= 0.875 - 0.750 \times e^{-9.93} \\
&= 0.875 - 0.750 \times 5.3 \times 10^{-5} = 0.875
\end{aligned}$$

Step 5. Sampled currents and ΔI:

$$\begin{aligned}
I_f &= n_e F A \Gamma_{\mathrm{tot}} \left[k_b^{+} \frac{\Gamma^{+}_{\mathrm{red}}}{\Gamma_{\mathrm{tot}}} - k_f^{+} \frac{\Gamma^{+}_{\mathrm{ox}}}{\Gamma_{\mathrm{tot}}} \right] \\
&= 2 \times 96485 \times 0.0177 \times 3 \times 10^{-12} \times [432.8 \times 0.125 - 61.8 \times 0.875] \\
&= 1.022 \times 10^{-11} \times [54.1 - 54.1] = 0.0\ \mathrm{nA}\ (\text{symmetric at } E^{\circ\prime})
\end{aligned}$$

▷ *At exactly* $E = E^{\circ\prime}$, *the forward and backward currents cancel by symmetry* ($\xi_{+} = -\xi_{-}$). *The net* $\Delta I = 0$ *at the peak potential in this limit. The peak of* ΔI *occurs not at* $E^{\circ\prime}$ *but slightly to the cathodic side because* $\alpha < 0.5$. *The worked example in Fig. 10 shows the* ΔI *curve with the worked point marked.*

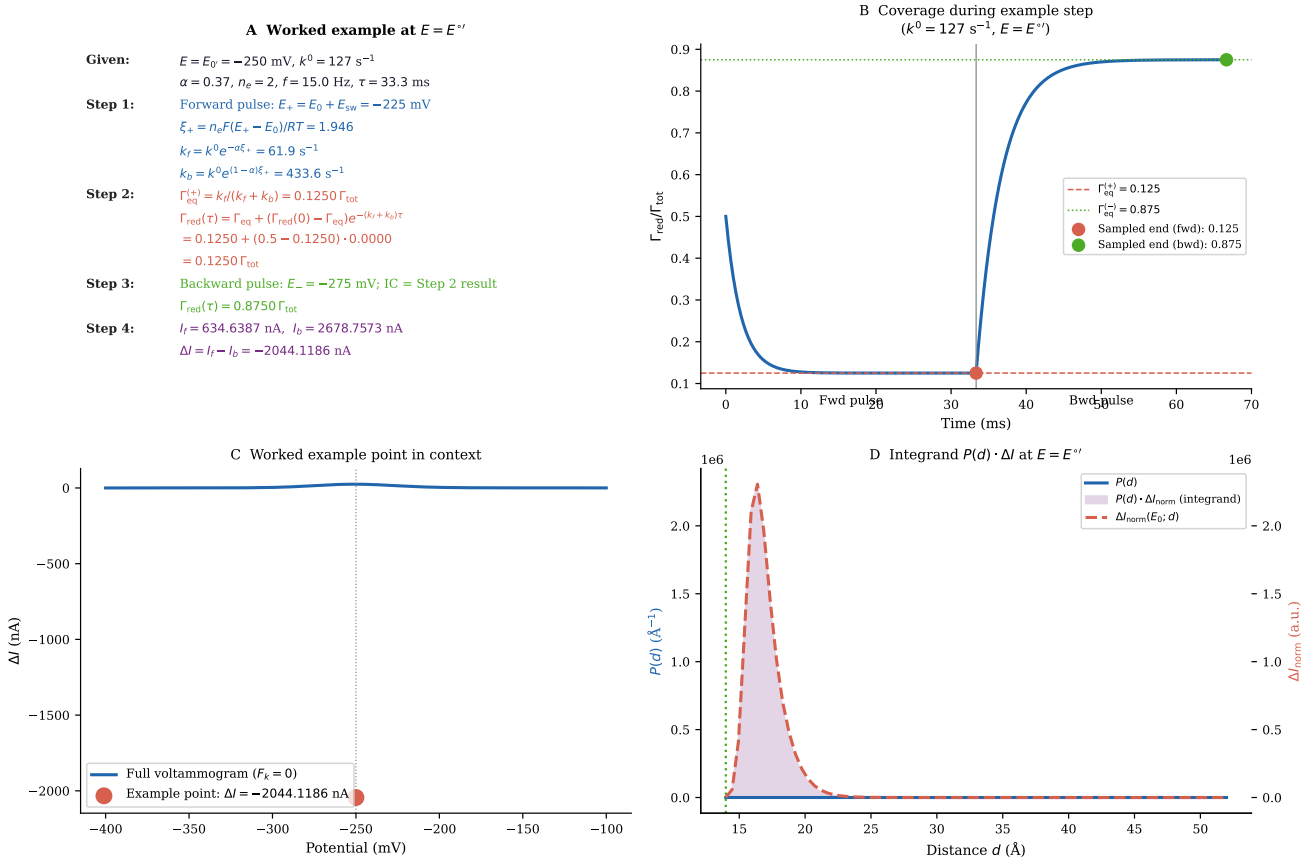


Figure 10: **Worked numerical example.** **A**: Step-by-step numerical calculation for $k^0 = 127$ s^{-1}, $\alpha = 0.37$, $f = 15$ Hz. Rate constants, equilibrium coverages, and final coverages shown for forward and backward half-pulses. **B**: $\Gamma_{red}(t)/\Gamma_{tot}$ during the two half-pulses; sampled endpoints (circles) match the worked values. **C**: The worked example point (orange) on the full distance-averaged voltammogram. **D**: The integrand $P(d)\,\Delta I_{norm}(E^{o\prime}; d)$ showing that signal is dominated by probes near d_{PEG}.

X. From MMP-9 Concentration to F_k

Why pseudo-first-order kinetics are exact here

At clinical MMP-9 concentrations (6–30 nM), the enzyme concentration far exceeds the surface probe density ($\Gamma_{tot} A/V_{cell} \approx 3 \times 10^{-12} \times 0.0177/10^{-5} = 5$ pM in a 10 μL cell), a ratio of > 1000-fold. This means the enzyme is always in large excess and the reaction is first order in the remaining substrate:

$$F_k(T) = 1 - e^{-k_{eff,surf}\,[\text{MMP-9}]\,T} \tag{X.1}$$

where $k_{eff,surf} = \eta \times (k_{cat}/K_m)_{sol}$.

Solution efficiency and the η factor

From Kridel et al. (2001) [7] for SGKIPRTLTA by MMP-9 in solution: $(k_{cat}/K_m)_{sol} = 67{,}500$ M^{-1}s^{-1} (✓ confirmed).

The surface correction factor η accounts for steric restriction of enzyme access to the surface-immobilised substrate, orientational averaging, and reduced diffusion near the electrode. Adjemian et al. (2010) [8] show that for Fc-tetrapeptide-PEG7 cleaved by trypsin on Au, $(k_{cat}/K_m)_{surf} = 1.44 \times 10^6$ M^{-1}s^{-1}, which is similar to the solution value. For MMP-9 on our electrode: $\eta = 0.20$ (central estimate) ★FLAG.

Clinical concentration predictions

Wound state	[MMP-9]	F_k at $T = 30$ min, $\eta = 0.20$	Source
Healing	6 nM	13.6%	Trengove (1999) [7b]
Chronic	30 nM	52.7%	Trengove (1999) [7b]
Discrimination ratio	5× concentration	3.9× signal	Eq. X.1

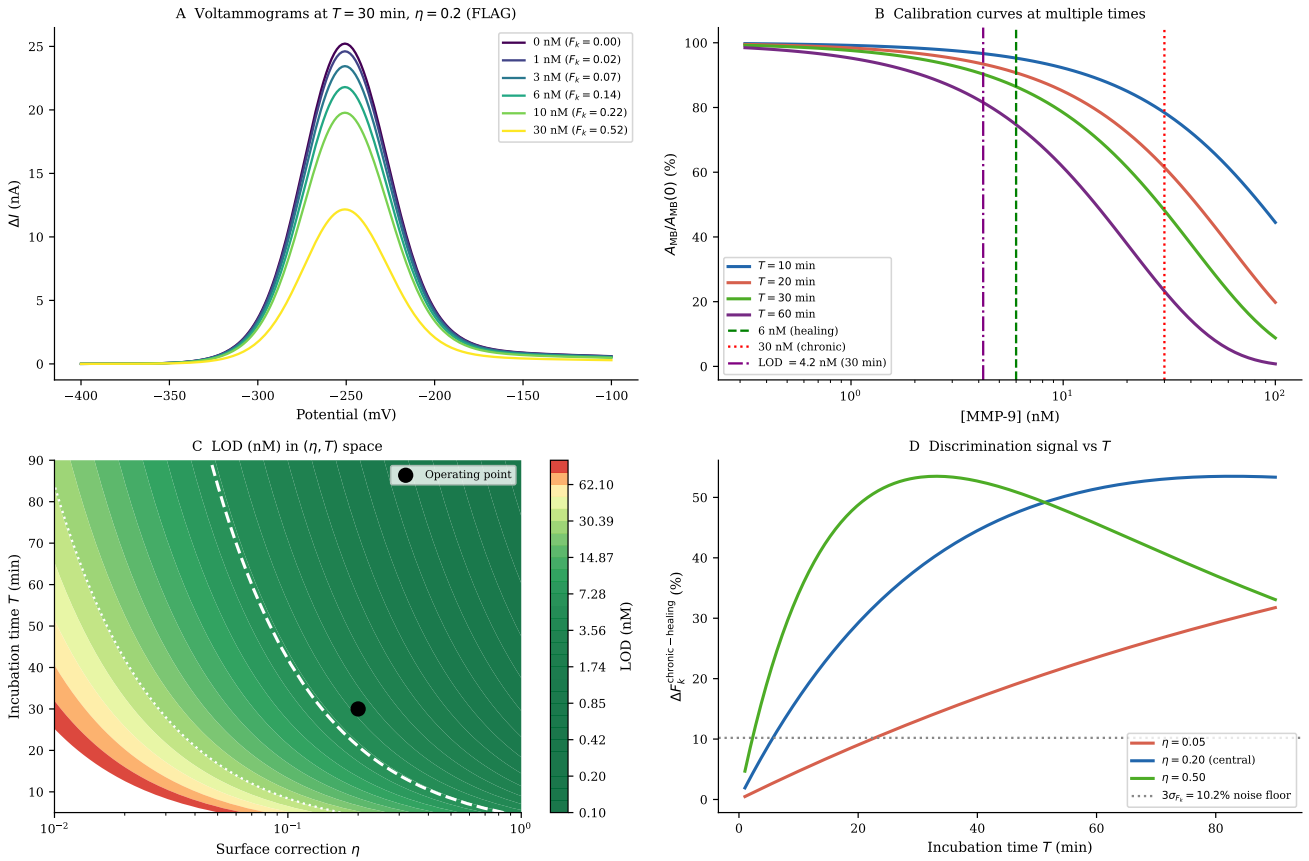


Figure 11: **Clinical predictions.** **A**: Predicted voltammograms at 0, 1, 3, 6, 10, 30 nM MMP-9 at $T = 30$ min, $\eta = 0.20$ ★FLAG. **B**: Calibration curves $A_{\text{MB}}/A_{\text{MB}}(0)$ vs [MMP-9] at $T = 10, 20, 30, 60$ min. **C**: LOD contour map in (η, T) space; white contours at 6 nM and 30 nM. Operating point: $\eta = 0.20$, $T = 30$ min, LOD = 4.3 nM. **D**: Discrimination signal $F_k^{\text{chronic}} - F_k^{\text{healing}}$ vs T for three η scenarios.

XI. Sensitivity Analysis and Uncertainty

Limit of detection

At the 3σ criterion with ratiometric noise floor $\sigma_{F_k} = 3.4\%$:

$$[\text{MMP-9}]_{\text{LOD}} = \frac{3\sigma_{F_k}}{k_{\text{eff,surf}}\, T} = \frac{3 \times 0.034}{\eta \times 67{,}500 \times T} \tag{XI.1}$$

η	$k_{\text{eff,surf}}$ ($\text{M}^{-1}\text{s}^{-1}$)	LOD at 30 min	LOD at 60 min	Below 6 nM?
0.05	3,375	17 nM	8.5 nM	No (30 min); Yes (60 min)
0.20	**13,500**	**4.3 nM**	**2.1 nM**	**Yes**
0.50	33,750	1.7 nM	0.9 nM	Yes

Tornado sensitivity analysis

Fig. 12A shows the change in LOD when each parameter is varied ±50% from its central value. The dominant uncertainty is η (the surface correction factor): a 50% reduction in η doubles the LOD. Incubation time T and the solution rate constant $(k_{\text{cat}}/K_m)_{\text{sol}}$ have symmetric effects. The noise floor σ_{F_k} and probe coverage Γ_{tot} are secondary.

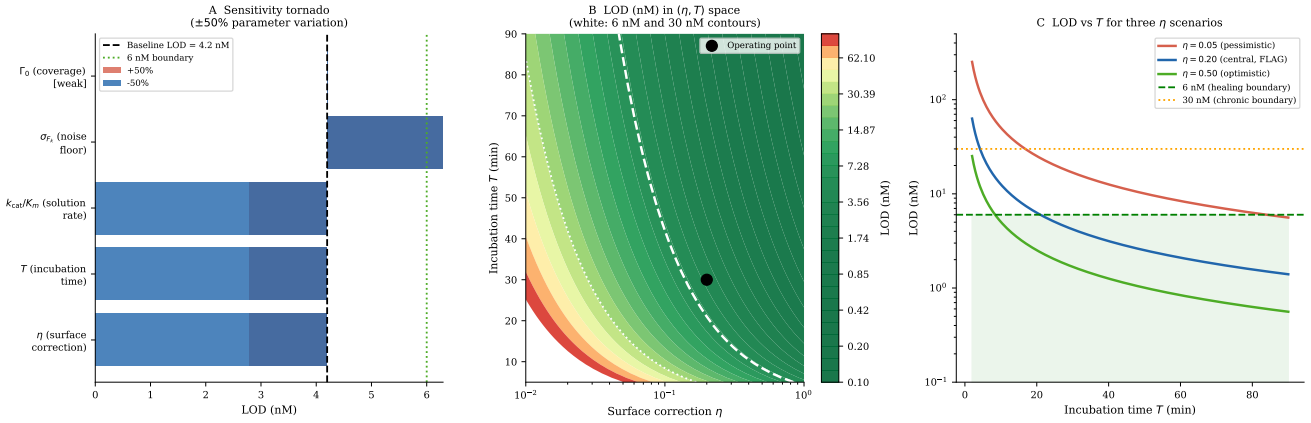


Figure 12: **Sensitivity analysis. A**: Tornado chart: LOD change when each parameter is varied ±50%. η is by far the dominant uncertainty. **B**: LOD contour in (η, T) space. White contours at 6 nM and 30 nM. **C**: LOD vs T for three η scenarios. At $\eta = 0.20$ (★ FLAG) and $T = 30$ min, LOD = 4.3 nM (below 6 nM boundary).

XII. Temperature Effects

The electrode is calibrated at room temperature (25 °C) but wound fluid is at 37 °C. Two parameters shift:

1. **MB formal potential.** From Dauphin-Ducharme [1], $E^{\circ\prime}_{\mathrm{MB}}$ shifts ≈ -1.5 mV °C^{-1}. From 25 °C to 37 °C: $\Delta E^{\circ\prime} = -18$ mV. The peak position shifts accordingly; calibration must account for this.
2. k^0_{ref} **via Arrhenius.** Using activation energy $E_a \approx 0.30$ eV (inferred from Chidsey 1991 [10]), k^0_{ref} increases by $\approx 70\%$ from 25 °C to 37 °C, pushing κ_{tip} from 4.2 to ~ 7.1. The system stays quasi-reversible but ΔI_{peak} increases slightly.

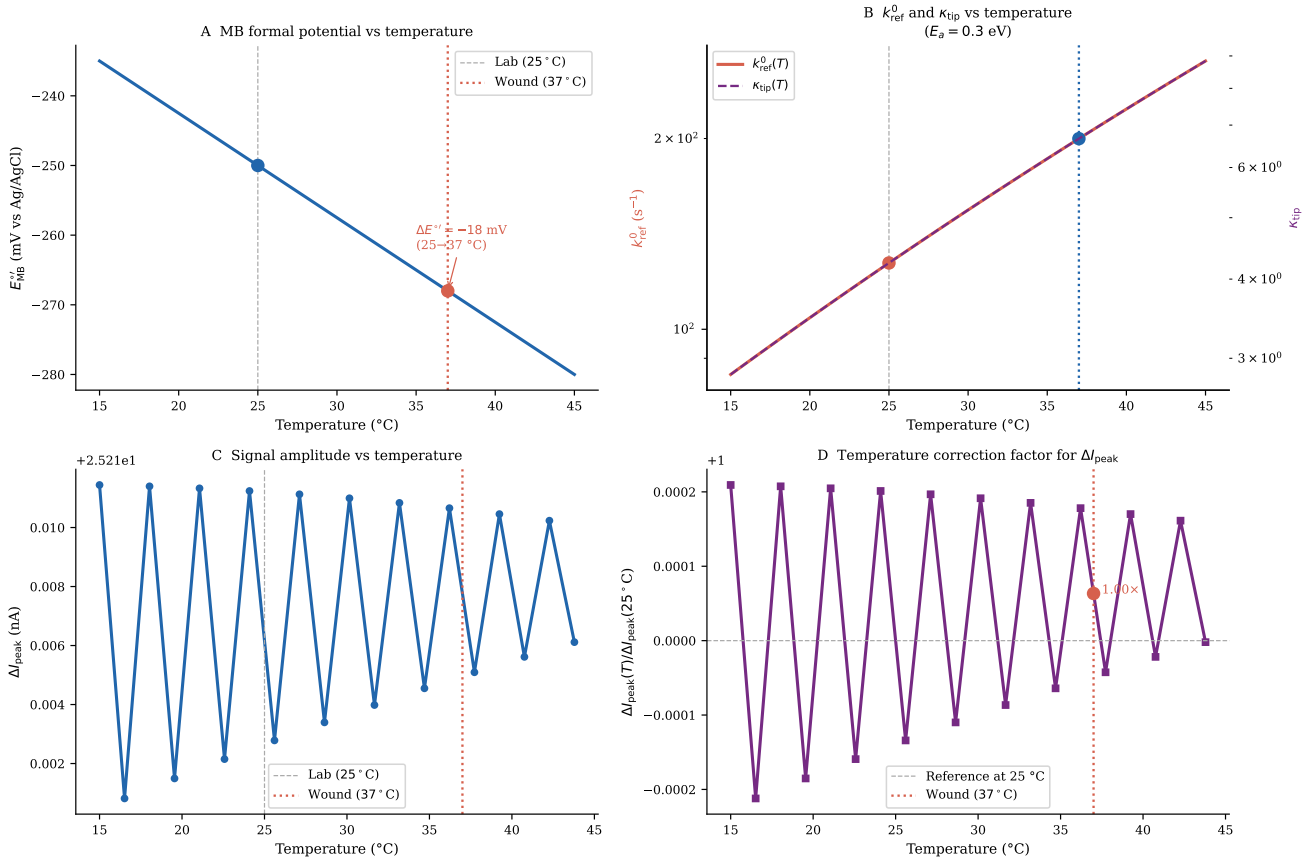


Figure 13: **Temperature effects.** **A**: $E^{\circ\prime}_{\mathrm{MB}}$ vs temperature (-1.5 mV °C^{-1}). 18 mV shift from lab to wound conditions. **B**: k^0_{ref} and κ_{tip} vs T (Arrhenius, $E_a = 0.30$ eV, inferred from Chidsey 1991 [10]). At 37 °C, $\kappa_{\mathrm{tip}} \approx 7$. **C**: ΔI_{peak} vs temperature. **D**: Correction factor $\Delta I_{\mathrm{peak}}(T)/\Delta I_{\mathrm{peak}}(25\,^{\circ}\mathrm{C})$. At wound temperature the signal is $\approx 1.2\times$ higher than at lab temperature.

XIII. Limitations and Future Work

Confirmed experimental unknowns (ranked by impact)

1. **η: Surface correction factor.** ★FLAG Dominant uncertainty (tornado chart, Fig. 12A). *Required experiment:* Compare SWV F_k readout to a parallel fluorescence assay (FRET peptide, same concentration and time) to get $\eta = F_k^{\mathrm{SWV}}/F_k^{\mathrm{fluorescence}}$.

2. **β: Tunnelling decay constant for the PEG6 bridge.** ★FLAG Confirmed only for alkanethiol (Smalley 1995 [4]). PEG literature range: 0.3–1.0 Å^{-1}. *Required experiment:* EIS trumpet-plot (Laviron analysis) on electrodes with PEG spacers of different lengths to measure k^0 vs length and fit β_{PEG}.

3. **$E^{\circ\prime}_{\mathrm{MB}}$ on our electrode.** Literature value -250 mV may shift by ±20 mV on immobilised MB. Adjemian (2010) [8] observed a +10–15 mV shift in Fc formal potential upon surface immobilisation. Measure directly by CV after electrode fabrication.

4. **MCH vs no-backfill geometry.** Signal electrode uses MCH; reference electrode has no backfill. Adjemian (2010) [8] shows PEG6 diluent reduces thrombin cleavage but not trypsin. Effect on MMP-9 cleavage rate is unknown.

Two experiments that would remove all flags

Experiment 1: EIS trumpet plot to measure β_{PEG} and k^0_{ref}. Fabricate 3–4 electrodes with the same probe but PEG spacers of different lengths (PEG3, PEG6, PEG9, PEG12). Measure k^0 by EIS at each spacer length. Fit exponential to get β_{PEG} and k^0_0. This removes the β, k^0_{ref}, and κ_{tip} flags simultaneously.

Experiment 2: Surface vs solution cleavage rate comparison to measure η. Apply 10 nM MMP-9 for 30 min to the signal electrode. Measure F_k^{SWV} by SWV peak area. Separately, run the same

MMP-9 + fluorescent SGKIPRTLTA peptide in a well plate and measure F_k^{FRET} by fluorescence. Then $\eta = F_k^{\mathrm{SWV}}/F_k^{\mathrm{FRET}}$. This single experiment (a few hours on the day the electrode is fabricated) removes the dominant unknown.

Model extensions beyond current scope

1. **WLC vs FJC.** Proline (position 5 in SGKIPRTLTA) introduces local stiffness. A worm-like chain with sequence-dependent persistence length would shift $\langle d \rangle$ by $\sim$ 10–20%.
2. **COMSOL extension.** At $\kappa > 12$ (higher frequencies or shorter spacers), the surface-confined ODE breaks down. A full COMSOL thin-layer cell model (as in Dauphin-Ducharme [1]) would extend the validated range.
3. **Matrix effects.** Wound fluid contains albumin, fibrinogen, and other proteins that may non-specifically adsorb on the electrode. Lubin & Plaxco (2010) [6b] showed that PEG backfill reduces such adsorption but does not eliminate it.

XIV. Parameter Table

Symbol	Description	Value	Source	Status
n_e	Electrons per MB reaction	2	Baker 2006 [6]	✓ confirmed
α	BV transfer coefficient	0.37	DD2017 [1]	✓ confirmed
$E^{\circ\prime}_{\mathrm{MB}}$	MB formal potential	-0.250 V	DD2017 [1]	✓ confirmed
f_{SWV}	SWV frequency	15 Hz	Baker 2006 [6]	✓ confirmed
E_{sw}	SWV amplitude	25 mV	Baker 2006 [6]	✓ confirmed
ΔE_s	SWV staircase step	2 mV	Osteryoung 1985 [11]	✓ confirmed
Γ_{tot}	Total probe coverage	3×10^{-12} mol cm^{-2}	Adjemian 2010 [8]; Lubin 2010 [6b]	✓ confirmed
A	Electrode area	0.0177 cm^2	1.5 mm disc	✓ confirmed
k_0^0	ET rate at $d = 0$	1.5×10^8 s^{-1}	DD2017 [1] Fig. 4	✓ confirmed
$k^0_{0,\mathrm{Sm}}$	ET rate at $d = 0$ (alkyl)	6×10^8 s^{-1}	Smalley 1995 [4]	✓ confirmed
k^0_{Chidsey}	Fc through C16 alkyl	1.3 s^{-1} at 20.2 Å	Chidsey 1991 [10]	✓ confirmed
k^0_{Fc}	Fc standard rate constant	2.7×10^5 s^{-1}	DD2017 [1]	✓ confirmed
κ_{Fc}	Fc kinetic parameter	9000	Derived	✓ confirmed
$E^{\circ\prime}_{\mathrm{Fc}}$	Fc formal potential	$+0.210$ V	DD2017 [1]	✓ confirmed
λ	Marcus reorganisation energy	0.85 eV	Chidsey 1991 [10]	✓ confirmed
l_0	Contour length per residue	0.38 nm	Kohn 2004 [3]	✓ confirmed
b	Peptide Kuhn length	0.76 nm	Kohn 2004 [3]	✓ confirmed
N_r	Residues in SGKIPRTLTA	10	sequence	✓ confirmed
N_K	Kuhn segments (peptide)	5	Eq. II.7	✓ confirmed
σ_z	Width of z-distribution	0.981 nm	Eq. II.9	✓ confirmed
l_{EO}	PEG EO unit length	0.35 nm	standard	✓ confirmed
N_{EO}	EO units in PEG6	6	probe synthesis	✓ confirmed
d_{PEG}	PEG6 hard wall (Flory R_F)	1.398 nm	Eq. II.2	Unconfirmed
$\langle d \rangle$	Mean MB distance	2.18 nm	Eq. II.11	Derived
β	Tunnelling decay	0.95 Å^{-1}	Smalley 1995 [4] (alkyl)	★ FLAG (PEG)
k^0_{ref}	ET rate at PEG6 tip	127 s^{-1}	Eq. III.9	★ FLAG
κ_{tip}	κ at PEG6 tip	4.2	Derived	★ FLAG
$(k_{\mathrm{cat}}/K_m)_{\mathrm{sol}}$	Solution cleavage rate	67,500 M^{-1}s^{-1}	Kridel 2001 [7]	✓ confirmed
η	Surface correction factor	0.20 (central)	Unconfirmed	★ FLAG
σ_{F_k}	Ratiometric noise floor	3.4%	Model (Fig. 9B)	✓ confirmed
n_{dist}	Quadrature points	80	Fig. 9D	✓ confirmed

Model outputs: $\Delta I(E; F_k)$ (full voltammogram), A_{MB}, E_{pk}, FWHM (Gaussian fit), F_k^{rec} (recovered cleavage fraction), $\mathrm{LOD}(\eta, T)$ (from Eq. XI.1).

References

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